

Empirical non-equivalence

A

912 genomes · 31,457 variable sites · 1,804 branches

**31,644 informative
site-edge comparisons**

30,817 (97.39%)

Fixed-exclusive + unambiguous
focal-edge substitution

675 (2.13%)

Unambiguous focal-edge substitution,
not fixed-exclusive

152 (0.48%)

Placement-ambiguous

**No fixed-exclusive comparison lacked
an unambiguous focal-edge substitution.**

**Why were 675 unambiguous focal-edge substitutions
not fixed-exclusive?**

**645 (95.56%)
derived nucleotide also observed
outside the focal clade**

30 (4.44%)
descendant clade not fixed

Clade A recurrent-state example

B

AP009378.1_4301132

